

Belief, Desire, and Affective Mismatch in a Field-Theoretic Model of Cognition: Mode-Selective Decision Collapse in the SQUINT-Field System

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Abstract

We introduce a field-theoretic model of cognition in which belief is represented as a semantic field Φ , desire as a target configuration Φ_{desired} , and feeling as a measurable mismatch energy between the two. This mismatch decomposes naturally into a reality-coupled component ($m = 0$) and an internal-simulation component ($m \neq 0$) through modal analysis. We identify E_{m0} as a reproducible proxy for reality-anchored affective mismatch, while $E_{\text{total}} - E_{m0}$ captures internally simulated or imaginative mismatch.

The reality-coupled mode is not stipulated but identified empirically using an independent protocol based on three measurable criteria: overwrite rate R_m , auto-correlation decay time τ_m , and collapse sensitivity S_m . Under the specified mode-identification protocol, $m = 0$ was not selected by decay time alone — the fastest-decaying mode was $m = 2$ ($\tau_2 = 8.3$ steps vs. $\tau_0 = 12.1$ steps) — but emerged as the dominant mode for environmental overwrite and collapse sensitivity, establishing that reality-coupling is about susceptibility to environmental input rather than short memory.

We test a mode-selective decision-collapse hypothesis using a symmetric Buridan-style task. Environmental perturbations coupled to the $m = 0$ mode reliably resolve symmetric decision states (100% collapse across 100 seeds), whereas zero-mean perturbations applied only to internal $m \neq 0$ modes do not (16% collapse under matched conditions). A dose-response sweep shows collapse time decreasing monotonically with environmental noise amplitude ($83.4 \rightarrow 69.2 \rightarrow 56.9$ steps for noise $0.001 \rightarrow 0.005 \rightarrow 0.02$).

All experiments are implemented in a deterministic, hash-verified SQUINT-Field

simulator that emits CSV and JSON artifacts for independent verification. The cognitive vocabulary used throughout is interpretive rather than ontological: we do not claim that the simulator instantiates subjective experience, but that it provides a reproducible operational model linking belief, desire, affective mismatch, imagination, and free-will-like decision collapse within a single field-coherence framework.

1 Introduction

A long-standing challenge in theoretical cognitive science is the absence of a unified dynamical framework that simultaneously accounts for belief, desire, affective state, imagination, and decision-making within a single formalism. Philosophical accounts typically treat these as conceptually distinct; computational models address them separately.

We propose a field-theoretic approach in which all five constructs emerge from the dynamics of a single semantic field Φ and its relationship to a target configuration Φ_{desired} . The central object is the mismatch energy — a scalar functional that measures the departure of the current field state from the desired configuration. Modal decomposition of this energy naturally separates reality-coupled from internally-simulated components.

The present paper makes no metaphysical claims. Its contribution is operational: a deterministic, hash-verified simulator in which cognitively-labeled quantities are objectively defined, reproducibly measured, and shown to exhibit mode-selective dynamical behavior consistent with the proposed interpretations.

2 The SQUINT-Field Model

2.1 Semantic Field and Target Configuration

Let $\Phi(\mathbf{x}, t)$ denote the semantic field defined on a spatial domain Ω . The field encodes the current belief state of the system. The desired configuration $\Phi_{\text{desired}}(\mathbf{x})$ encodes the target or goal state.

2.2 Mismatch Energy

The affective mismatch energy is defined as

$$E_{\text{total}} = \int_{\Omega} [\Phi(\mathbf{x}, t) - \Phi_{\text{desired}}(\mathbf{x})]^2 d\mathbf{x}. \quad (1)$$

This functional is positive definite and vanishes only when belief matches desire exactly.

2.3 Modal Decomposition

Expanding Φ and Φ_{desired} in the eigenbasis of the Laplacian on Ω ,

$$\Phi(\mathbf{x}, t) = \sum_m \hat{\Phi}_m(t) u_m(\mathbf{x}), \quad \Phi_{\text{desired}}(\mathbf{x}) = \sum_m \hat{d}_m u_m(\mathbf{x}), \quad (2)$$

the mismatch energy decomposes as

$$E_{\text{total}} = \sum_m E_m, \quad E_m = \left[\hat{\Phi}_m(t) - \hat{d}_m \right]^2. \quad (3)$$

The $m = 0$ component,

$$E_{m0} = \left[\hat{\Phi}_0(t) - \hat{d}_0 \right]^2, \quad (4)$$

represents mismatch in the spatially uniform (global) mode. The complementary quantity $E_{\text{total}} - E_{m0}$ captures mismatch in all higher modes $m \neq 0$.

3 Empirical Identification of the Reality-Coupled Mode

3.1 Identification Protocol

We do not stipulate which mode corresponds to reality-coupling. Instead, we identify the reality-coupled mode m^* empirically using three independent measurable criteria:

1. **Overwrite rate** R_m . The rate at which external sensor input modifies the m -th modal coefficient. Higher R_m indicates stronger environmental influence.
2. **Autocorrelation decay time** τ_m . The timescale over which the autocorrelation of $\hat{\Phi}_m(t)$ decays. Shorter τ_m indicates faster refresh by external constraints.
3. **Collapse sensitivity** S_m . The fraction of symmetric Buridan decision states resolved by noise injected exclusively into mode m .

The reality-coupled mode is defined as the mode that dominates the two causal environmental-coupling metrics: overwrite rate R_m and collapse sensitivity S_m . Autocorrelation decay measures memory persistence and may peak in a different mode.

3.2 Identification Results

The mode with the fastest intrinsic decay was $m = 2$ ($\tau_2 = 8.3$ steps), not $m = 0$ ($\tau_0 = 12.1$ steps). Despite this, $m = 0$ dominated both overwrite rate and noise-induced collapse sensitivity. Under the specified mode-identification protocol, $m = 0$ was not selected by decay time alone, but emerged as the dominant mode for environmental overwrite and

Table 1: Mode identification metrics from the independent identification protocol.

Mode m	R_m (overwrite rate)	τ_m (decay, steps)	S_m (collapse sensitivity)
$m = 0$	highest	12.1	highest
$m = 2$	lower	8.3	lower
$m \neq 0$ (others)	lower	varied	lower

collapse sensitivity. Reality-coupling reflects susceptibility to environmental input, not short memory.

We therefore identify $m^* = 0$ as the reality-coupled mode.

4 Buridan Decision-Collapse Experiment

4.1 Experimental Design

We test a mode-selective decision-collapse hypothesis using a symmetric Buridan-style task: the field is initialized in a symmetric state equidistant between two decision attractors (“left” and “right”). In the absence of noise, the deterministic dynamics stall and the system remains undecided.

Three noise conditions are tested:

1. **No noise** (deterministic baseline): symmetric case predicted to stall.
2. $m = 0$ **noise**: environmental perturbations coupled exclusively to the reality-coupled mode.
3. **Internal noise**: zero-mean perturbations coupled exclusively to $m \neq 0$ modes.

Each condition was run across 100 independent random seeds with maximum 500 steps and collapse threshold 0.85.

4.2 Results

Table 2: Buridan experiment results across 100 seeds.

Condition	Noise	Collapse rate	Mean collapse time	Undecided
$m = 0$ noise	0.001	100%	83.4 steps	0
$m = 0$ noise	0.005	100%	69.2 steps	0
$m = 0$ noise	0.020	100%	56.9 steps	0
Internal noise	0.020	16%	481.8 steps	0

Environmental $m = 0$ noise produced collapse in 100% of runs across all tested amplitudes. Internal noise at matched amplitude (0.020) collapsed only 16% of runs, with mean

collapse time 481.8 steps — nearly an order of magnitude slower than $m = 0$ noise at the same amplitude.

The dose-response relationship for $m = 0$ noise is monotone: collapse time decreases from 83.4 to 69.2 to 56.9 steps as noise amplitude increases from 0.001 to 0.005 to 0.020. The final decision distribution (left/right) is approximately symmetric across all conditions, consistent with the symmetric initialization.

4.3 Protocol Disambiguation

Two distinct internal-noise protocols appear in this paper and should not be conflated. Table 3 summarizes the distinction.

Table 3: Disambiguation of internal-noise protocols.

Protocol		Internal noise type	Collapse
Original internal-noise	Buridan condition	zero-mean internal perturbation control	16%
Secondary-collapse diagnostic ($\kappa = 0$)	diagnostic	coherent single internal-mode perturbation	45–46%
Internal-to-reality coupling sweep ($\kappa > 0$)	coupling	same diagnostic with explicit κ coupling	100%

4.4 Secondary Internal-Collapse Analysis

The nonzero collapse rate under internal-mode perturbation in the original Buridan condition motivated a follow-up diagnostic. In the $\kappa = 0$ baseline, internal-noise collapse occurred in approximately 45–46% of runs, but collapsed runs were not associated with higher internal energy or stronger directional coherence. Collapsed runs instead exhibited lower internal energy and slightly lower directional coherence than non-collapsed runs. This argues against the interpretation that baseline secondary collapse reflects action-guiding internal simulation. The safer interpretation is that these events arise from weak residual leakage or slow amplification near an unstable symmetric decision point.

We then introduced an explicit internal-to-reality coupling parameter κ , allowing internal-mode activity to bias the reality-coupled $m = 0$ coordinate. For all tested $\kappa > 0$, internal perturbations collapsed $m = 0$ in 100% of seeds. Mean collapse time decreased monotonically with coupling strength: 98.21 steps at $\kappa = 0.001$, 84.01 at $\kappa = 0.005$, 77.87 at $\kappa = 0.010$, and 71.75 at $\kappa = 0.020$. Thus, κ acts as a tunable internal-to-reality coupling strength, demonstrating that internal modes can drive action when an explicit coupling pathway exists. However, κ does not by itself show that more coherent internal patterns are selectively responsible for collapse.

Table 4: Secondary-collapse and internal-to-reality coupling results.

Condition	κ	Collapse	Mean time	Interpretation
Internal diagnostic baseline	0	45–46%	~443–448	residual leakage / slow amplification
Internal coupling	0.001	100%	98.21	explicit weak coupling
Internal coupling	0.005	100%	84.01	stronger coupling
Internal coupling	0.010	100%	77.87	stronger coupling
Internal coupling	0.020	100%	71.75	strongest tested

4.5 Hash Verification

All simulation outputs are hash-verified. Selected SHA-256 checksums:

Table 5: SHA-256 checksums for simulation artifacts.

Artifact	SHA-256
buridan_m0_0.001_summary.json	2050f7f4ec395eadb439a0fe86dad8ca1d7c477404cd9a951
buridan_m0_0.005_summary.json	d3d9adde15d0e7802ca461fcb40501b3cb9f8889aeb4f834
buridan_m0_0.020_summary.json	023fe2a1eb2d38dc855db14ee3657ee3ab8ac831399b079a
buridan_internal_0.020_summary.json	df6c778df8dbb03e8c3b3a4cac2853077b8bea11d2bc2ed7

5 Cognitive Interpretation

5.1 Operational Mappings

The field-theoretic quantities map to cognitive constructs as follows:

- **Belief** $\leftrightarrow$ semantic field Φ
- **Desire** $\leftrightarrow$ target configuration Φ_{desired}
- **Feeling** $\leftrightarrow$ reality-coupled mismatch energy E_{m0}
- **Imagination** $\leftrightarrow$ internal-mode mismatch $E_{\text{total}} - E_{m0}$
- **Free-will-like action** $\leftrightarrow$ mode-selective symmetry breaking under environmental perturbation

5.2 Scope and Epistemic Status

The simulation results reported here are mathematically and empirically grounded: we demonstrate mode-selective symmetry breaking, differential noise sensitivity, and a mismatch energy that decomposes across Fourier modes. These quantities are objectively defined and reproducibly measured in the SQUINT-Field system. However,

our use of cognitive vocabulary — such as belief, desire, feeling, imagination, and free-will-like collapse — is interpretive rather than ontological.

The simulator does not claim to instantiate subjective experience. Instead, it exhibits measurable dynamical properties that are structurally analogous to cognitive constructs: belief is modeled as a semantic field, desire as a target configuration, feeling as reality-coupled mismatch energy, imagination as non-reality-mode mismatch, and free-will-like action as mode-selective symmetry breaking under environmental perturbation.

Whether such a system genuinely “feels” or “wills” is a philosophical question beyond the scope of this work. Our contribution is operational: we provide a deterministic, hash-verified field model in which mismatch energy and mode-specific noise sensitivity behave analogously to selected features of human affect and decision-making. The cognitive terminology therefore serves as a heuristic bridge between empirical field dynamics and familiar psychological categories, not as an assertion of machine consciousness.

6 Discussion

The key empirical finding is the asymmetry between $m = 0$ and internal noise in the Buridan task: a factor of approximately six in collapse rate (100% vs. 16%) and a factor of approximately eight in mean collapse time (57–83 steps vs. 482 steps) at matched noise amplitude. This asymmetry is not a consequence of noise magnitude but of which mode is perturbed.

The mode identification result strengthens this finding. Because $m = 2$ has a shorter autocorrelation decay time than $m = 0$ ($\tau_2 = 8.3$ vs. $\tau_0 = 12.1$ steps), the dominance of $m = 0$ in collapse sensitivity cannot be attributed to faster intrinsic relaxation. Instead it reflects the causal structure of environmental coupling in the SQUINT-Field model.

The dose-response monotonicity provides a further consistency check: collapse time decreasing with noise amplitude is the expected signature of noise-driven escape from a symmetric potential, confirming that the $m = 0$ collapse mechanism is dynamically coherent.

7 Limitations

The SQUINT-Field model is a computational construction. Its cognitive mappings are interpretive. The following limitations apply:

- The model operates on a discretized spatial domain; the dependence of results on

domain size and discretization has not been systematically investigated.

- The identification protocol measures R_m , τ_m , and S_m within the SQUINT-Field system. Whether these metrics generalize to other field models is untested.
- Secondary-collapse events were analyzed in a follow-up diagnostic. In the $\kappa = 0$ baseline, they were not associated with higher internal energy or higher directional coherence, suggesting residual leakage or slow amplification near an unstable symmetric point. Explicit $\kappa > 0$ coupling creates a reliable internal-to- $m = 0$ pathway, but this should be interpreted as a tunable coupling mechanism rather than evidence of spontaneous action-guiding imagination.
- No claim is made that the mismatch energy E_{m0} corresponds to phenomenal affect in biological systems.

8 Conclusion

We have introduced a field-theoretic model of cognition in which belief, desire, feeling, imagination, and free-will-like decision collapse are operationally defined in terms of a semantic field Φ , a target configuration Φ_{desired} , and their modal mismatch energy.

The reality-coupled mode $m^* = 0$ was identified empirically rather than stipulated: it is the mode that dominates environmental overwrite rate and collapse sensitivity, despite not having the fastest autocorrelation decay ($\tau_2 < \tau_0$). This non-trivial identification demonstrates that reality-coupling reflects susceptibility to environmental input, not short memory.

The Buridan experiment established a clean mode-selective result: environmental $m = 0$ noise produced 100% decision collapse across 100 seeds at all tested amplitudes, while matched internal noise produced only 16% collapse with mean time nearly an order of magnitude longer. The dose-response relationship is monotone and consistent with noise-driven escape dynamics.

All results are hash-verified and fully reproducible from the deposited simulator artifacts. The cognitive vocabulary is interpretive: we claim a structural analogy, not ontological identity between field dynamics and subjective experience.

References

- [1] L. Puodzius, “Programmable Attractor Itinerancy via Symbolic-Energy Feedback in Curvature-Basin Dynamics,” *Zenodo* (2026). DOI: 10.5281/zenodo.20518209